

1. Reverse a String

```
function reverseString(string) {  
    console.log(string.split("").reverse().join(""));  
};  
reverseString("Automation");    //Output: 'noitamotuA'
```

2. Check for Palindrome

```
function palindrome(data) {  
    let  
result=(data==data.split("").reverse().join(""))?"Yes it  
is a palindrome":"No it is not a palindrome";  
    return result;  
};  
console.log(palindrome("233"));  
                                //Output: 'No it is not a  
palindrome'  
console.log(palindrome("121"));  
                                //Output: 'Yes it is a palindrome'
```

3. Fibonacci Series

```
function toGetFibonacci(n) {  
    let fibseries=[0,1];  
    for(let i=2;i<=n;i++){  
        fibseries.push(fibseries[i-1]+fibseries[i-2]);  
    }  
    return fibseries.slice(0,n);  
}  
console.log(toGetFibonacci(4)); //Output: [ 0, 1, 1, 2 ]
```

4. Factorial of a Number

```
function toGetFactorial(n,factorial=1){
    if(n<0) return "Factorial is undefined for negative
integers";
    for(let i=1;i<=n;i++){
        factorial*=i;
    }
    return factorial;
};
console.log(toGetFactorial(5)); //Output: 120
```

5. Prime Number Check

```
function isPrime(n) {
    if (n <= 1) return false;
    for (let i = 2; i <= Math.sqrt(n); i++) {
        if (n % i === 0) {
            return false;
        }
    }
    return true;
}
console.log(isPrime(16)); //Output: false
console.log(isPrime(3)); //Output: true
```

6. Count Vowels and Consonants

```
function countVowelsANdConsonants( word,vowels="aeiou",
    vowelscount=0, consonantscount=0){
for(char of word){
    if(/[a-zA-Z]/.test(char)){
        if(vowels.includes(char))
            vowelscount+=1;
        else
            consonantscount+=1;
    }
}
return `vowelscount:${vowelscount}
consonantscount:${consonantscount}`;
}
console.log(countVowelsANdConsonants("abeiou23kol56#s"));
//Output: vowelscount:6 consonantscount:4
```

7. Sort an Array

```
function sortAnArray(numbers){
return numbers.sort();
}
console.log(sortAnArray([1,5,10])); //Output: [ 1, 10, 5 ]
```

8. Merge Two Arrays

```
function MergingArrays(array1,array2){  
    return array1.concat(array2);  
};  
console.log(MergingArrays([1,2,3,4],[5,6,7,8]));  
//Output: [  
           1, 2, 3, 4,  
           5, 6, 7, 8  
           ]
```

9. Find the Largest Element in an Array

```
function toFindLargestElement(array){  
    return Math.max(...array);  
};  
console.log(toFindLargestElement([1,2,3,4,5]));  
//Output: 55
```

10. Remove Duplicates from an Array

```
function removeDuplicates(array){  
    let  
    filteredarray=array.filter((value,index,numbers)=>{return  
    numbers.indexOf(value)==index});  
    return filteredarray;  
}  
console.log(removeDuplicates([1,2,4,2,1]));  
//Output: [ 1, 2, 4 ]
```

11. Check if a Number is Armstrong

```
function toCheckArmstrong(number) {
    let array=number.toString().split("").map(Number);
    let
total=array.reduce((sum,value)=>sum+Math.pow(value,array.l
ength),0);
    return result=total==number?"Is Armstrong":"Not
Armstrong";
};
console.log(toCheckArmstrong(153));
//Output: "is Armstrong"
console.log(toCheckArmstrong(121));
//Output: "not Armstrong"
```

12. Reverse a Number

```
function toReverseNumber(number) {
    let
reversednum=Number(Math.abs(number).toString().split("").r
everse().join(""));
    return number<0?-reversednum:reversednum;
};
console.log(toReverseNumber(12345)); //Output: 54321
console.log(toReverseNumber(-12345)); //Output: -54321
```

13. Calculate GCD of Two Numbers

```
function toGetGCD(a,b) {  
    while(b!=0) {  
        let temp=b;  
        b=a%b;  
        a=temp;  
    }  
    return a;  
};  
console.log(toGetGCD(8,0)); //Output: 8
```

14. Check for Anagram

```
function checkAnagram(word1,word2) {  
    let  
    string1=word1.replace(/\s/g,'').toLowerCase().split('').so  
    rt().join("");  
    let  
    string2=word2.replace(/\s/g,'').toLowerCase().split('').so  
    rt().join("");  
    return string1==string2?"Is Anagram":"Not  
Anagram";  
};  
console.log(checkAnagram("Elbow","below"));  
//Output: "Is Anagram"  
console.log(checkAnagram("hello","hell"));  
//Output: "Not Anagram"
```

15. Count the Number of Digits in a Number

```
function toCountDigits(number) {  
    return (Math.abs(number).toString()).length;  
};  
console.log(toCountDigits(123));    //Output: 3  
console.log(toCountDigits(-123));  //Output: 3
```

16. Print the Prime Numbers in a Range

```
function primeNumber(n,m) {  
    let primenum=[];  
    for(i=n;i<=m;i++){  
        if(i<2) continue;  
        let isPrime=true;  
        for(j=2;j<=Math.sqrt(i);j++){  
            if(i%j==0){  
                isPrime=false;  
                break;  
            }  
        }  
        if(isPrime) primenum.push(i);  
    }  
    return primenum;  
}  
console.log(primeNumber(10,20)); //Output: [ 11, 13, 17,  
19 ]
```

17. Find the Second Largest Element in an Array

```
function toFindSecondLargest(array) {  
    return  
    array.length<2?"There is no largest element":  
    [...new Set(array)].sort((a,b)=>b-a)[1];  
};  
console.log(toFindSecondLargest([2,5,3,8,7,9,10]));  
//Output: 9
```

18. Print the Pascal's Triangle

```
function pascalTriangle(rows) {  
    const triangle=[];  
    for(let i=0;i<rows;i++) {  
        triangle[i]=[1];  
        for(let j=1;j<i;j++) {  
  
triangle[i][j]=triangle[i-1][j-1]+triangle[i-1][j];  
        }  
        if (i>0) triangle[i].push(1);  
    }  
    for(let row of triangle){  
        console.log(...row);  
    }  
};  
pascalTriangle(5);    //Output: 1
```

```
1 1  
1 2 1  
1 3 3 1  
1 4 6 4 1
```

19. Swap Two Numbers

```
function swapNumbers(a,b) {  
    [a,b]=[b,a]  
    return [a,b];  
};  
console.log(swapNumbers(1,2));    //Output:[ 2, 1 ]
```

20. Find the Missing Number in an Array

```
function toFindMissingNumber(array) {  
    let missnum=[];  
    let set=[...new Set(array)];  
    for(i=1;i<=Math.max(...array);i++) {  
        if(!set.includes(i))  
            missnum.push(i);  
    }  
    return missnum;  
};  
console.log(toFindMissingNumber([1,3,4,5,7,9]));  
//Output: [ 2, 6, 8 ]
```

21. Convert Decimal to Binary

```
function decimalToBinary(n) {  
    let reminder=[];  
    while(n>0) {  
        reminder.push(n%2);  
        n=Math.floor(n/2);  
    }  
    return Number(reminder.reverse().join(""));  
}  
console.log(decimalToBinary(13));    //Output: 1101
```

22. Check for Perfect Number

```
function checkForPerfectNum(n, sum=0) {
    for(i=1; i<n; i++) {
        if(n%i==0)
            sum=sum+i;
    }
    return sum==n?"It is a Perfect number":"It is not a
Perfect number";
};
console.log(checkForPerfectNum(28));
//Output: "It is a Perfect number"
console.log(checkForPerfectNum(7));
//Output: "It is not a Perfect number"
```

23. Implementing a Simple Calculator

```
function simpleCalculator(a,b,sign){
    if ((typeof a === "number") && (typeof b ===
"number")) {
        switch (sign) {
            case "+":
                return a + b;
            case "-":
                return a - b;
            case "*":
                return a * b;
            case "/":
                return a / b;
        }
    }
    return "unknown value";
};
console.log(simpleCalculator(1,2,"+")) //Output: 3
```

24. Find the Sum of Digits of a Number

```
function sumOfDigits(number, sum=0) {  
    return  
    sum=number.toString().split("").reduce((add, value) => add+Number(value), 0);  
};  
console.log(sumOfDigits(12345)); //Output: 15
```

25. Find the Length of a String

```
function length(string) {  
    return string.length;  
};  
console.log(length("kulsum")); //Output: 6
```

26. Check if a String is Empty

```
function emptyString(string) {  
    if(string=="") return "string is empty";  
};  
console.log(emptyString("")); //Output: string is empty
```

27. Count the Occurrences of a Character in a String

```
function count(string, char) {  
    return (string.match(char)).length;  
};  
console.log(count("karnataka", /a/g)); //Output: 4
```

28. Find the First Non-Repeated Character in a String

```
function nonRepeatChar(string) {  
    for(char of string){  
        let regex=new RegExp(char,'gi');  
        if((string.match(regex)).length==1) return  
char;  
    };  
    return null;  
};  
console.log(nonRepeatChar("MalaYalam")) //Output: Y
```

29. Remove All Whitespaces from a String

```
function removeWhite(string){  
    return string.replace(/\s/g, '');  
};  
console.log(removeWhite("a uto m obile"));  
//Output: automobile
```

30. Find the Common Elements in Two Arrays

```
function commonElements(array1,array2){  
    return  
array1.filter((value)=>array2.includes(value));  
};  
console.log(commonElements([1,2,3],[3,2,4,5]));  
//Output: [ 2, 3 ]
```

31. Find the Factorial of a Number using Recursion

```
function factorial(n) {  
    if (n === 1) return 1;  
    return n * factorial(n - 1);  
console.log(factorial(5));           //Output: 120
```

32. Generate Random Numbers

```
function randomNumber() {  
    return Math.round(Math.random()*10000);  
}  
console.log(randomNumber());        //Output: 3170
```

33. Check if a Year is Leap Year

```
function leapYear(year) {  
    return ((year%4==0 && year%100!=0) ||  
(year%400==0)?true:false);  
};  
console.log(leapYear(1600));        //Output: true  
console.log(leapYear(1600));        //Output: false
```

34. Find the Sum of First N Natural Numbers

```
function naturalNumber(n, sum=0) {  
    for(i=1; i<=n; i++) {  
        sum+=i  
    }  
    return sum;  
};  
console.log(naturalNumber(3))       //Output: 6
```

35. Implement a Simple Login System

1. Username

- Starts with `USR` in upper case
- Having -
- Til max length 4 that digit

2. Write a regex for password

- Should starts with `alphaa-z`
- One upper case should be there
- 4 digit should be there
- Special symbol `#+-` should

```
function validateLogin(username,password) {
    let nameregex=/^[A-Z].*\-\d{0,4}.*$/g;
    let
passregex=/^[a-z] (?!.*[A-Z]) (?!.*\d{4,}) (?!.*[#\+\-]) .*$ /g
;

    let isusernamevalid= nameregex.test(username);
    let ispasswordvalid=passregex.test(password);
    if(isusernamevalid && ispasswordvalid){
        console.log("Login successful!");
    }else{
        console.log("Login failed:");
        if(!isusernamevalid) console.log("-
Invalid username format.");
        if(!ispasswordvalid) console.log("-
Invalid password format.");
    }
};

validateLogin("Kulsum211-ga","kulsum1234#U");
//Output: "Login successful!"
validateLogin("Kulsum211ga","kulsum1234#U");
//Output: Login failed:
- Invalid username format.
```

36. Check if a String Contains Another String

```
function stringCheck(string, substring) {  
    return  
    string.toLowerCase().includes(substring.toLowerCase());  
};  
console.log(stringCheck("JavaScript", "script"));  
                //Output: true  
console.log(stringCheck("JavaScript", "hell"));  
                //Output: false
```

37. Find the Maximum Occurring Character in a String

```
function toGetMaxChar(string) {  
    let maxlength=0;  
    let maxchar;  
    for(char of string) {  
        let regex=new RegExp(char, 'gi');  
        let maxarray=string.match(regex);  
        if (maxarray.length>maxlength) {  
            maxlength=maxarray.length;  
            maxchar=maxarray[0];  
        }  
    }  
    return maxchar;  
};  
console.log(toGetMaxChar("hello")); //Output: l
```

38. Implementing Bubble Sort

```
function bubbleSort(array) {  
    let n=array.length;  
    let swap;  
    for(let i=0; i<n-1; i++){  
        for(let j=0; j< n-1-i; j++){  
            if(array[j] > array[j+1]){  
                swap=array[j];  
                array[j]=array[j+1];  
                array[j+1]=swap;  
            }  
        }  
    }  
    return array;  
};  
console.log(bubbleSort([5,2,4,3,1]))  
//Output: [ 1, 2, 3, 4, 5 ]
```

39. Implementing Selection Sort

```
function selectionSort(array) {  
    let n=array.length;  
    let swap;  
    for(i=0;i<n-1;i++) {  
        let minIndex=i;  
        for (j=i+1;j<n;j++) {  
            if(array[j]< array[minIndex]) {  
                minIndex=j;  
            }  
        }  
        if (minIndex!==i) {  
            swap=array[i];  
            array[i]=array[minIndex];  
            array[minIndex]=swap;  
        }  
    }  
    return array;  
};  
console.log(selectionSort([5,2,4,3,1]));  
//Output: [ 1, 2, 3, 4, 5 ]
```